Adroit Technologies Protocol Driver

Name	BACnet Ethernet UDP/IP
DLL	D_BACNET

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1. Overview

This driver makes use of the BACnet Stack library supplied by sourceforge.com.

The driver supports these functions at the moment:

ReadProperty
ReadPropertyMultiple
WriteProperty

Not all devices support ReadPropertyMultiple.
1. For reading we will always try to do a ReadPropertyMultiple and if it fails the driver will revert to a
2. ReadProperty call instead. ReadProperty is much slower and has a limitation of only supporting 255
3. Items that can be scanned individually.

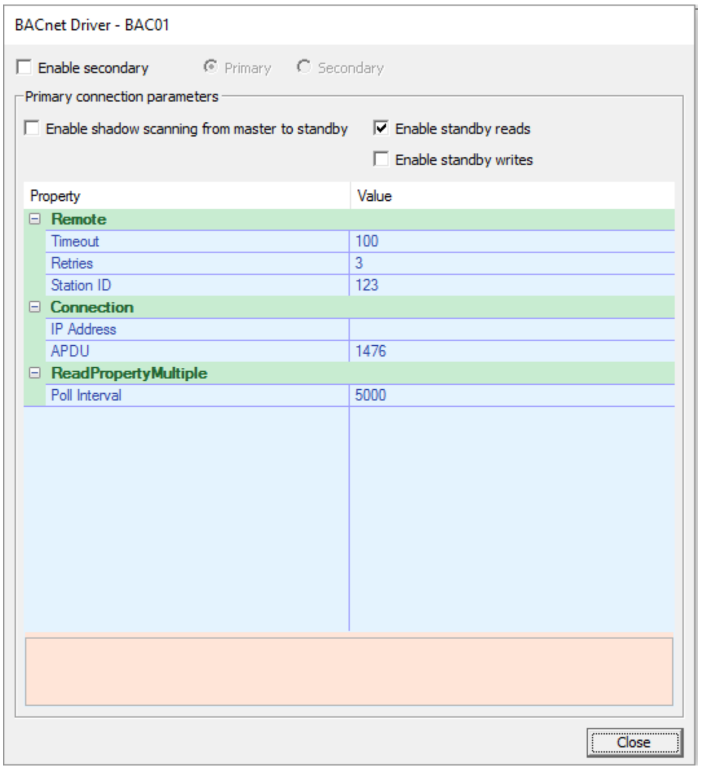
The driver also supports Ethernet connection through a WhoIs function call.

NOTE: Please ensure broadcasting is enabled on all routers and internal network between driver and BACnet devices to ensure successful communications. Also only UDP is used so ensure UDP is enabled through any firewalls. UDP port 47808 is used.

2. Wiring

Not Applicable.

3. Device Configuration



3.1. Driver Details

Enable Secondary – Allows enabling of redundant BACnet device connections such that in case of a failure a switchover will occur. (Default = OFF)

Primary – Allows configuration setting of the Primary BACnet device.

Secondary – Allows configuration setting of the Secondary BACnet device.

3.1.1. BACnet Device Details

Enable shadow scanning from master to standby – If enabled will stop any communication to the BACnet device when the Agent server is in standby mode. By default we assume parallel scanning (default = OFF)

Enable standby reads – Allows reading from BACnet device when agent server is in standby mode. (Default = ON)

Enable standby writes – Allows writing to BACnet device when agent server is in standby mode. (Default = OFF)

Timeout – Time in milliseconds to wait for a response from the BACnet device. (Default = 1000 ms)

Retries – The maximum number of retries to attempt communication to the BACnet device before ensuring a complete failure. (Default = 3)

Station ID – The BACnet device Station ID. This is only required and comes into working when the actual IP Address of the unit is unknown or will change dynamically.
IP Address – The static IP Address of the BACnet device. If this is specified the Station ID is ignored as connection happens purely based on IP and UDP Port then.

APDU – The maximum size in bytes that can be transferred over Ethernet. 1495 bytes is the maximum allowed on a single network packet and then some devices may implement segmentation of packets. As not all devices support segmentation, the driver in itself will segment the packets before sending it.

Poll Interval – The time in milliseconds at which all the items will be requested from the device.

PLEASE NOTE

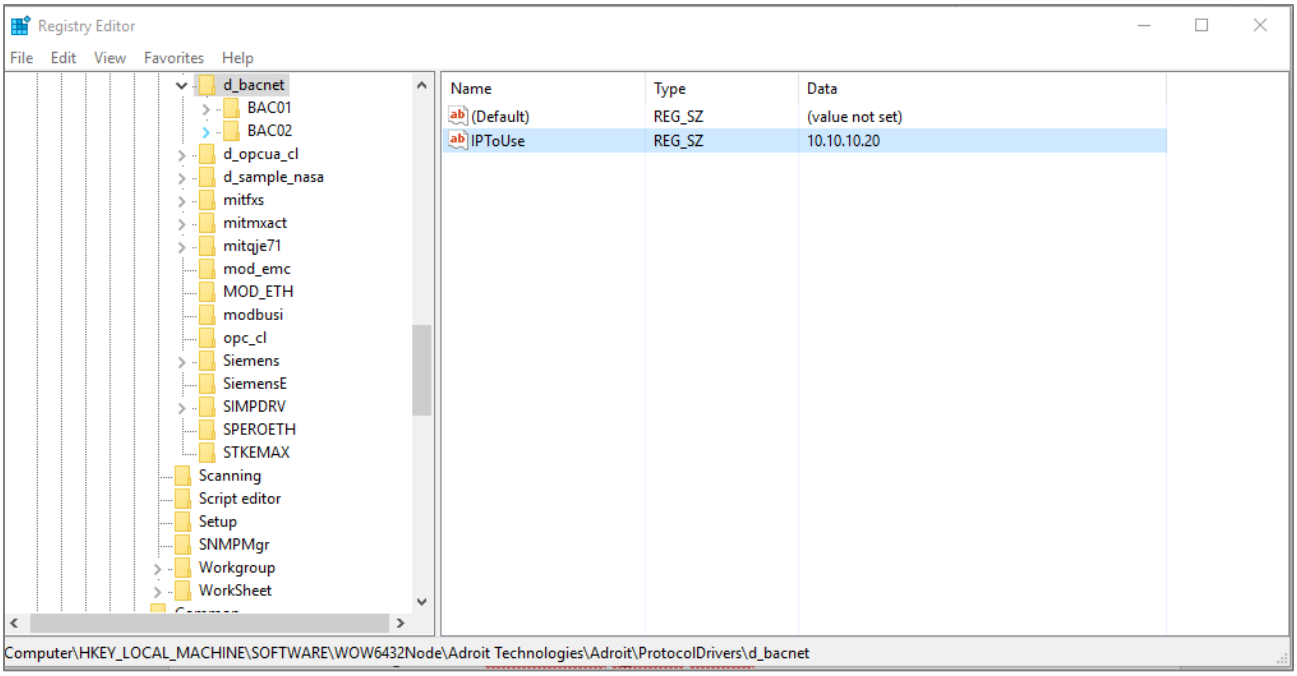
NB – If using the Station ID for WhoIs command the BACnet stack makes use of the very first NIC on your PC to send broadcast messages. In some cases, this specific NIC chosen is not the same one that the BACnet devices is running on and hence no communications. To overcome this scenario, a global registry key called "IPToUse" is created in the Drivers registry area with the default used NIC's IP address in it. Here you can change the IP Address to the current NIC you want to use to access your BACnet devices with. See image below.

The registry key for 32 Bit O/S:

'HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\d_bacnet\IPToUse'

The registry key for 64 Bit O/S:

'HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\Adroit Technologies\Adroit\ProtocolDrivers\d_bacnet\IPToUse'



4. Supported Addresses

Scan Addresses takes the form of 3/4 digits separated with a dot as in 0.1.77.1!

The 4 digits are represent as follows:
DIGIT 1: The BACnet Object Type. For example ANALOG_INPUT object = 0. Full list of objects shown below!

DIGIT 2: The BACnet Object Instance.

DIGIT 3: The BACnet Property. For example OBJECT_NAME = 77. Full list of properties shown below.

DIGIT 4: The BACnet Property Index of Array. This not required and only for properties, that has a value in an array!

4.1. BACnet Objects List:

OBJECT_ANALOG_INPUT	0
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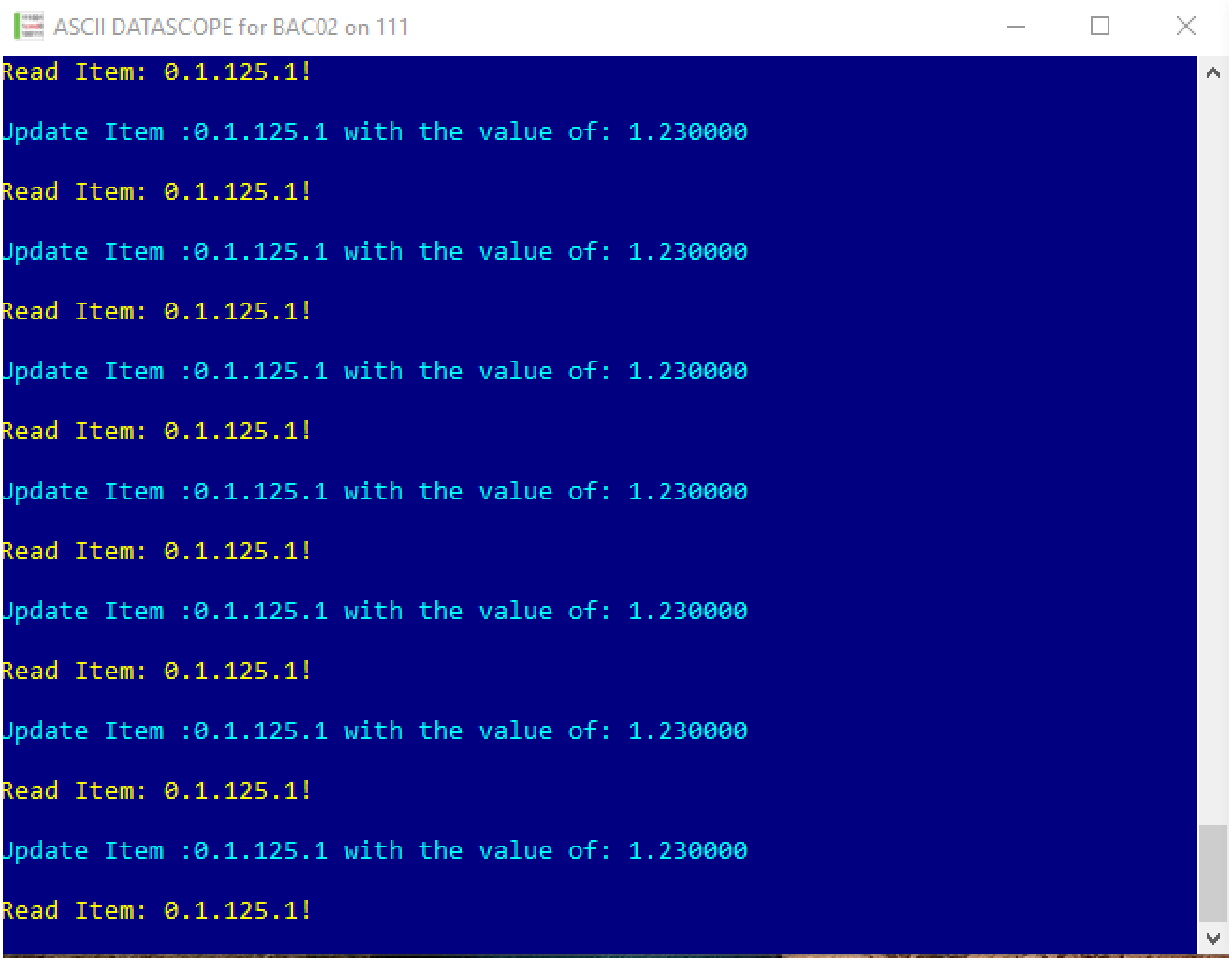
5. Protocol Definition and Considerations



6. General Notes and Troubleshooting

6.1. Contact Details

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6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback (adroittechnologiesautomation.com)

7. Appendix A – Mitsubishi AE-200/AE-50/EW-50 Aircon Control Unit

These Mitsubishi Air conditioning control units are used as a gateway to communicate to multiple AIRCON units, each having its own unique ID.

As an example of how the addressing relates to the object ID, let's take the object ID "BO_01xx01" that relates to an address of 4.10101.85, where the BO is a Binary Output Object and therefore the number 4 is used, the second part of the object ID namely 01xx01 is translated into 010101 where the xx is replaced with the actual AIRCON units unique ID. In this case using 01, and the leading 0 is then removed to get to 10101. Now all that needs to be added is the property. Refer to the property table above and specify and property you require, in this specific case we are referencing the "Current Value" property which is the 85 number. The full address then for object ID BO_01xx01 is translated to BACnet address 4.10101.85.

Below is the BACnet address mapping when using these controllers.

MITSUBISHI AIRCON CONTROL UNIT					
Object	**Object Type**	**Object ID**	**BACnet**	**Agent**	**Values**
On/Off Control	Binary Output (4)	BO_01xx01	4.10101.85	Digital	INACTIVE(0): OFF ACTIVE(1): ON
On/Off State	Binary Input (3)	BI_01xx02	3.10102.85	Digital	INACTIVE(0): OFF ACTIVE(1): ON
Alarm Signal	Binary Input (3)	BI_01xx03	3.10103.85	Digital	INACTIVE(0): Normal ACTIVE(1): Error
Error Code	Multi-State Input (13)	MI_01xx04	13.10104.85	Stringlist	01: Normal 02: Other errors 03: Refrigeration system fault 04: Water system error 05: Air system error 06: Electronic system error 07: Sensor fault 08: Communication error 09: System error
Operational Mode Control	Multi-State Output (14)	MO_01xx05	14.10105.85	Stringlist	01: Cool 02: Heat 03: Fan 04: Auto 05: Dry 06: Setback
Operational Mode State	Multi-State Input (13)	MI_01xx06	13.10106.85	Stringlist	01: Cool 02: Heat 03: Fan 04: Auto 05: Dry 06: Setback
Fan Speed Control	Multi-State Output (14)	MO_01xx07	14.10107.85	Stringlist	01: Low 02: High 03: Mid 2 04: Mid 1 05: Auto
Fan Speed State	Multi-State Input (13)	MI_01xx08	13.10108.85	Stringlist	01: Low 02: High 03: Mid 2 04: Mid 1 05: Auto
Room Temperature	Analog Input (0)	AI_01xx09	0.10109.85	Analog	ºF/ºC (32ºF–199ºF/0.0ºC–99.0ºC)
Set Temperature	Analog Value (2)	AV_01xx10	2.10110.85	Analog	ºF/ºC (32ºF–199ºF/0.0ºC–99.0ºC)

Filter Sign	Binary Input (3)	BI_01xx11	3.10111.85	Digital	INACTIVE(0): OFF ACTIVE(1): ON
Filter Sign Reset	Binary Value (5)	BV_01xx12	5.10112.85	Digital	INACTIVE(0): Reset ACTIVE(1): Void
Prohibition On/Off	Binary Value (5)	BV_01xx13	5.10113.85	Digital	INACTIVE(0): Permit ACTIVE(1): Prohibit
Prohibition Mode	Binary Value (5)	BV_01xx14	5.10114.85	Digital	INACTIVE(0): Permit ACTIVE(1): Prohibit
Prohibition Filter Sign Reset	Binary Value (5)	BV_01xx15	5.10115.85	Digital	INACTIVE(0): Permit ACTIVE(1): Prohibit
Prohibition Set Temperature	Binary Value (5)	BV_01xx16	5.10116.85	Digital	INACTIVE(0): Permit ACTIVE(1): Prohibit
Prohibition Fan Speed	Binary Value (5)	BV_01xx17	5.10117.85	Digital	INACTIVE(0): Permit ACTIVE(1): Prohibit
M-NET Communication State	Binary Input (3)	BI_01xx20	3.10120.85	Digital	INACTIVE(0): Normal ACTIVE(1): Error
Individual System Forced Off	Binary Value (5)	BV_01xx21	5.10121.85	Digital	INACTIVE(0): Reset ACTIVE(1): Execute
Collective System Forced Off	Binary Value (5)	BV_019921	5.19921.85	Digital	INACTIVE(0): Reset ACTIVE(1): Execute
Air Direction Control	Multi-State Output (14)	MO_01xx22	14.10122.85	Stringlist	01: Horizontal 02: Downblow 60% 03: Downblow 80% 04: Downblow 100% 05: Swing
Air Direction State	Multi-State Input (13)	MI_01xx23	13.10123.85	Stringlist	01: Horizontal 02: Downblow 60% 03: Downblow 80% 04: Downblow 100% 05: Swing
Set Cool Temperature	Analog Value (2)	AV_01xx24	2.10124.85	Analog	ºF/ºC (32ºF–199ºF/0.0ºC– 99.0ºC)
Set Heat Temperature	Analog Value (2)	AV_01xx25	2.10125.85	Analog	ºF/ºC (32ºF–199ºF/0.0ºC–99.0ºC)
Set Auto Temperature	Analog Value (2)	AV_01xx26	2.10126.85	Analog	ºF/ºC (32ºF–199ºF/0.0ºC–99.0ºC)
Set High Limit Setback Temperature	Analog Value (2)	AV_01xx27	2.10127.85	Analog	ºF/ºC (32ºF–199ºF/0.0ºC–99.0ºC)
Set Low Limit Setback Temperature	Analog Value (2)	AV_01xx28	2.10128.85	Analog	ºF/ºC (32ºF–199ºF/0.0ºC–99.0ºC)
Ventilation Mode Control	Multi-State Output (14)	MO_01xx35	14.10135.85	Stringlist	01: Heat Recovery 02: Bypass 03: Auto
Ventilation Mode State	Multi-State Input (13)	MI_01xx36	13.10136.85	Stringlist	01: Heat Recovery 02: Bypass 03: Auto
Air to Water mode Control	Multi-State Output (14)	MO_01xx37	14.10137.85	Stringlist	01: Heating 02: Heating ECO 03: Hot Water 04: Anti-freeze 05: Cooling
Air to Water mode State	Multi-State Input (13)	MI_01xx38	13.10138.85	Stringlist	01: Heating 02: Heating ECO 03: Hot Water 04: Anti-freeze 05: Cooling
Night Purge State	Binary Input (3)	BI_01xx46	3.10146.85	Digital	INACTIVE(0): OFF ACTIVE(1): ON
Thermo On/Off State	Binary Input (3)	BI_01xx47	3.10147.85	Digital	INACTIVE(0): Thermo OFF ACTIVE(1): Thermo ON
System Alarm Signal	Binary Input (3)	BI_010048	5.10048.85	Digital	INACTIVE(0): Normal ACTIVE(1): Error
Error Code Detail	Analog Input (0)	AI_01xx49	0.10149.85	Integer	Normal: 8000 Error: Error code (4 digits)
External Heat Source State	Binary Input (3)	BI_01xx50	3.10150.85	Digital	INACTIVE(0): Thermo OFF ACTIVE(1): Thermo ON

8. Revisions

8.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1		K. Robinson	Changes to format, updated image resources.	D. Jordaan
2.0				
3.0				
4.0				
5.0				
6.0				

8.2. DLL Revision

Rev.	Date	Changes
1	02/05/2017	First Release of Driver.
	15/11/2018	Included ReadPropertyMultiple and static IP Addressing.
	08/01/2020	Added Appendix A with Mitsubishi Aircon Unit Controller Mappings.
31	04/01/2023	Added Switch connection, and primary / secondary connection code added. Missing QualityChanged check to item updates added, Disabled logical comres support/